Coefficient array storage correspondence 2026-10-31

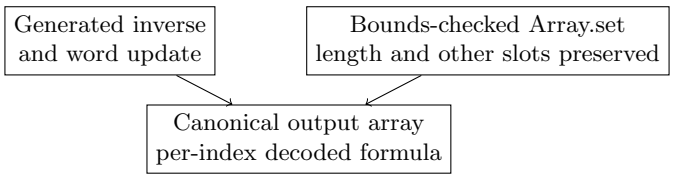
T-1163. Producer proof; independent source-fidelity review pending.

Bounds-checked array replacement preserves length, writes the selected slot and leaves every other slot unchanged. Canonical residues remain canonical when the replacement is canonical.

For prime p with $1 < p < 2^{31}$, canonical coefficients and parameters, nonzero scale s, and $0 \le j \le k \le 5$, the composed generated-inverse update gives

$$\operatorname{decode}(\mathrm{out}'_i) = \operatorname{decode}(\mathrm{out}_i) + \begin{cases} c\binom{k}{j} a^j b^{k-j} s^{-1} & i = j, \\ 0 & i \neq j. \end{cases}$$

The source index condition $j \le k <$ length proves the array access is in bounds.



Seven main declarations and eight controls have separate named axiom readbacks. Controls cover first/middle/last replacements, preserved neighbours, empty-array size, an excluded out-of-bounds index, canonical zeros and an actual generated coefficient update. The first attempt needed explicit `Array.getElem_set` rewriting for untouched slots; the corrected attempt passed. The first controls attempt needed an explicit three-index split in place of an unavailable tactic. A second controls attempt needed the literal size and proof-dependent hypotheses handled explicitly. All native receipts and warnings remain recorded.

This is one finite-array mutation composed with checked arithmetic. Full nested-loop accumulation, trailing-zero trimming and fixed-size divisor copying remain the next bindings. Lean array semantics are a source storage model, not compiled-Rust equivalence.

Source bindings: `projective_law_gate.rs` transform; preceding `Loop.lean`, `WordCoefficient.lean`, `Inversion.lean`. Primary reference: Lean Array.set semantics. No novelty claim for standard array or field identities.

Native exit codes, unchanged source/dependency bindings and axiom readback accompany this PDF, editable SVG and structured status. Existing compiled dependencies remain in the trust boundary. No scientific run, timing or cost ratio.